

Task 1: Technical Requirements

A food-ordering application requires:

- Web/mobile application servers
- Database servers
- Object storage for restaurant and food images
- Secure payment processing
- API services
- Load balancing
- CDN
- Authentication and IAM
- Monitoring and logging
- Backup and disaster recovery
- High availability
- Auto-scaling during peak periods

Task 2: Public vs Private vs Hybrid Cloud

Factor	Public Cloud	Private Cloud	Hybrid Cloud
Cost	Lower initial cost	Higher infrastructure cost	Medium to high
Security	Strong controls but requires correct configuration	Greater infrastructure control	Flexible security controls
Scalability	Excellent	More limited	Excellent
Maintenance	Lower infrastructure maintenance	Higher maintenance	Medium/high complexity
Traffic spikes	Very suitable	Requires additional capacity	Very suitable

Updates	Easier managed infrastructure	Organization manages more infrastructure	Requires coordination
---------	-------------------------------	--	-----------------------

Task 3: Recommended Model

Recommendation: Public Cloud

I would recommend the public cloud for a Zomato-style food-ordering application because it provides high scalability and reduces the need for the company to maintain physical infrastructure. During peak meal times or special events, cloud resources can scale to handle increased traffic. Public cloud services also provide managed databases, storage, networking, monitoring and security services that can simplify application management.

Task 4: Hybrid Cloud During a Huge Traffic Spike

Advantage

A hybrid cloud can use public-cloud resources to handle sudden increases in traffic while keeping selected sensitive workloads in private infrastructure.

Disadvantage

Managing communication, security and data synchronization between private and public environments becomes more complex. Incorrect configuration can also create additional security risks.

Task 5: When to Use Public, Private and Hybrid Clouds

Public Cloud

Best when an application needs:

- High scalability

- Fast deployment
- Lower initial infrastructure costs
- Global reach

Examples: Consumer applications such as BookMyShow or large-scale e-commerce platforms.

Private Cloud

Best when an organization requires:

- Greater infrastructure control
- Strict security requirements
- Specialized compliance requirements
- Dedicated infrastructure

Examples: Organizations handling highly sensitive workloads may choose private infrastructure for selected systems.

Hybrid Cloud

Best when an organization needs:

- Both public-cloud scalability and private infrastructure
- Flexible workload placement
- Controlled handling of sensitive workloads
- Extra capacity during traffic spikes

Examples: Large organizations such as financial services companies may use hybrid strategies for selected workloads.